

Dongfang Zhao, Ph.D.

Assistant Professor
Department of Computer Science
University of Nevada, Reno

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Status

Male, married, U.S. citizen

Research Areas

High-performance computing, databases, artificial intelligence, computer security

Professional Preparation

2005 Diploma, Statistics, Katholieke Universiteit Leuven, Belgium
2008 M.Sc., Computer Science, Emory University, Atlanta, Georgia
2015 Ph.D., Computer Science, Illinois Institute of Technology, Chicago
2017 Postdoc, Computer Science, University of Washington, Seattle

Appointments

2017–Now Assistant Professor of Computer Science, University of Nevada, Reno (UNR)
2018–Now Lead Software Developer, Moka Blox LLC., Davis, California
2018–2020 Visiting Professor of Computer Science, University of California, Davis
2016–2017 Research Associate, University of Washington, Seattle
2015 Research Intern, IBM Almaden Research Center, San Jose, California
2014 Research Intern, Argonne National Laboratory, Chicago, Illinois
2013 Research Intern, Pacific Northwest National Laboratory, Seattle, Washington
2012–2015 Research Assistant, Illinois Institute of Technology, Chicago
2009–2011 Software Developer, Epic Systems, Madison, Wisconsin
2008–2009 Software Developer, IDMI Systems, Atlanta, Georgia

Teaching

Cryptography and Blockchain	University of Nevada CS 705	2023–now
Distributed Computing	University of Nevada CS 791	2020–now
Database Management Systems	University of Nevada CS 457	2018–now

Services

2021–Now	Editorial Review Board	IEEE Transactions on Distributed & Parallel Systems
2022	Grant Review Panelist	National Science Foundation (2×)
2021	Faculty Search Committee	Department of Computer Science, UNR
2019	Organizing Committee	ACM/IFIP International Middleware Conference
2018	Grant Review Panelist	U.S. Department of Energy
2018	Grant Review Panelist	National Science Foundation

Awards

2022	Federal Research Grant, NSF 2112345, \$255,916, PI at University of Nevada “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, DOE DE-SC0020455, \$200,000, PI at University of Nevada “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2019	Industrial Research Grant, Amazon Web Services, \$30,000
2019	Industrial Research Grant, Google Cloud Platform, \$15,000
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2017	Industrial Research Grant, Microsoft Azure, \$20,000
2017	Industrial Research Grant, Amazon Web Services, \$10,000
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000
2007	Graduate Student Fellowship, Oak Ridge Institute for Science and Education, \$85,000

Selected Publications

Book chapters

- [1] **Dongfang Zhao**: Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. Handbook on Blockchain, Springer Optimization and its Applications. 2022.
- [2] **Dongfang Zhao**, Akash Mahakode, Sandip Lakshminarasiah, Ioan Raicu: High-Performance Storage Support for Scientific Big Data Applications on the Cloud. Resource Management for Big Data Platforms 2016: 147-170

Journals

([†]Student advised by **Dongfang Zhao**)

- [1] [†]Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data, doi: 10.1109/TB-DATA.2020.3035522.
- [2] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig: Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing. 8(2): 635-646 (2020)
- [3] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu: Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data 3(1): 107-117 (2017)
- [4] Tonglin Li, Xiaobing Zhou, Ke Wang, **Dongfang Zhao**, Iman Sadooghi, Zhao Zhang, Ioan Raicu: A convergence of key-value storage systems from clouds to supercomputers. Concurrency and Computation: Practice and Experience. 28(1): 44-69 (2016)
- [5] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, Ioan Raicu: Exploiting multi-cores for efficient interchange of large messages in distributed systems. Concurrency and Computation: Practice and Experience. 28(13): 3568-3585 (2016)
- [6] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, Ioan Raicu: Toward high-performance key-value stores through GPU encoding and locality-aware encoding. Journal of Parallel and Distributed Computing. 96: 27-37 (2016)
- [7] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, Zhiling Lan: I/O-aware bandwidth allocation for petascale computing systems. Parallel Computing. 58: 107-116 (2016)
- [8] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu: Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems. 27(6): 1824-1837 (2016)
- [9] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu: Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. IEEE Transactions on Service Computing 9(1): 96-109 (2016)
- [10] **Dongfang Zhao**, Li Yang: Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence. 31(1): 86-98 (2009)

Conferences

([†]Student advised by **Dongfang Zhao**)

- [1] [†]Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [2] [†]Olamide Timothy Tawose, [†]Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022. Direct acceptance without revision
- [3] [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [4] [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021. Direct acceptance without revision
- [5] [†]Xinying Wang, [†]Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**: HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020. Oral presentation
- [6] [†]Xinying Wang, [†]Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019. Best paper award
- [7] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan: Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [8] [†]Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**: Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018. Best student paper award
- [9] [†]Jiayao Wang, [†]Abdullah Al-Mamun, Tonglin Li, Linhua Jiang, **Dongfang Zhao**: Toward Performant and Energy-efficient Queries in Three-tier Wireless Sensor Networks. Proceedings of the 47th International Conference on Parallel Processing (ICPP), 2018.
- [10] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad: Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. Proceedings of the VLDB Endowment (VLDB). 10(11): 1226-1237 (2017)
- [11] Zhou Zhou, Xu Yang, **Dongfang Zhao**, Paul Rich, Wei Tang, Jia Wang, Zhiling Lan: I/O-Aware Batch Scheduling for Petascale Computing Systems. IEEE International Conference on Cluster Computing (CLUSTER), 2015.
- [12] **Dongfang Zhao**, Kan Qiao, Ioan Raicu: HyCache+: Towards Scalable High-Performance Caching Middleware for Parallel File Systems. The 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID), 2014.

- [13] **Dongfang Zhao**, Kent Burlingame, Corentin Debains, Pedro Alvarez-Tabio, Ioan Raicu: Towards high-performance and cost-effective distributed storage systems with information dispersal algorithms. IEEE International Conference on Cluster Computing (CLUSTER), 2013.
- [14] **Dongfang Zhao**, Chen Shou, Tanu Malik, Ioan Raicu: Distributed data provenance for large-scale data-intensive computing. IEEE International Conference on Cluster Computing (CLUSTER), 2013.
- [15] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu: ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. The 27th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2013.
- [16] **Dongfang Zhao**, Li Yang: Incremental Construction of Neighborhood Graphs for Nonlinear Dimensionality Reduction. The 18th International Conf. on Pattern Recognition (ICPR), 2006.